

Input image I

TRAIN semantic banks

TRAIN noun / verb lemmas

CLIP text encoder Fixed banks

Learned semantic aggregation

Pn, Pv: 150 × 512

Pn

Fusion stages share this structure; their parameters are separate.

patches

CLIP ViT-B/16 Visual encoder

ROI pooling Projection to 512

V

Boxes B

Object fusion 1

Object fusion 2

T1

Semantic selection

K,V

V1

Self-attention

Q

Cross-attention

Learned gate

Residual / Norm / FFN included

Vo

Object prompt attention Masked + residual

Vo′

Token prefix Embedding + position

Image-specific prompts

UPT detector

Object / relation phrases

CLIP text encoder Text embeddings to / tr

Object / relation slots MLPs: text + Vo + boxes

So

Transformer decoder ×3 Parallel cross-attention + gated fusion

YOLO detector

Pv

Relation fusion

Vr

Sr

Relation prompt attention Masked + residual

Vr′

Vocabulary head + softmax Beam search: 5

ROI links + masks

UPT boxes

UPT–YOLO ROI matching Class + IoU ≥ 0.5

Slots use original Vo. Masks restrict updates to linked regions.

YOLO boxes B

Five candidate captions

Weighted reranking 0.3D + 0.4C + 0.3O

Final caption

D: decoder likelihood
C: frozen CLIP image–text alignment
O: detector-based object consistency

∗ Frozen Trainable

Training: cross-entropy → SCST + cross-entropy